Image Processing and Medical Imaging

The Main Elements of The Digital Image Processing

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What is the image processing?

- Image processing, analysis, and machine vision is a very important part of computer science.
- Vision allows humans to understand the world surrounding them. So we can consider the vision as an important task for AI.
- The computer vision is a very difficult task for a computer.

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What is the image processing?

- On the one hand, to transform the pictures to better form for human understanding.
- On the other hand, to transform the pictures to better form for recognition by computer.
- Image processing and patter recognition, computer graphics are opposite tasks.
- What is the difference among optical image processing, computer image processing, digital image processing.

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History

- 1921 Sending image from London to New York through submarine cable. Problem with the noise.
- 1964 Processing photos sent by Ranger 7 about the Moon. Problem with the noise.
- 1965 Chromosome-classification by computer in the first issue of the first scientific journal on image processing.
- 1980 General problems, general computers.

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Why is computer vision difficult?

- **Loss of information in 3D to 2D.** A typical image capture device such as a camera can see the visual scene form only one viewpoint.
- **Interpretation.** Very difficult to analyze a visual scene.
- **Noise** is inherently present in each measurement in the real world.
- **Too much data.** An A4 sheet of paper scanned binary at 300 dpi at 8 bits per pixel corresponds to 8.5Mbyte.

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Why is computer vision difficult?

- **Brightness measured** in the image is given by complicated image formation physics. The radiance depends on the irradiance, the observer's position, the surface local geometry, and the surface reflectance properties.
- **Local window vs. global view.** It is often very difficult to interpret an image if it is seen only locally.

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Two levels of image analysis

- **Low-level processing** (very little knowledge about the content, data representation is comprised of original image)
- **High-level processing** is basen on knowledge, goals to understand the visual scene. High-level data are usually expressed in symbolic form.

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Image representation

- Mathematical **model**: a signal is a function on some variables (continuous, discrete, digital).
- The **quality** of a digital image: spatial, spectral, radiometric, time resolution.
- Image **digitization** means that the function *f(x,y)* is sampled into a matrix with *M* rows and *N* columns. Image quantization assigns to each continuous sample an integer.

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Sampling and quantization

- A continuous image is digitized at **sampling points**. Their geometric relation is called the **grid**.
- The transition between continuous values of the image function (brightness) and its digital equivalent is called **quantization**.

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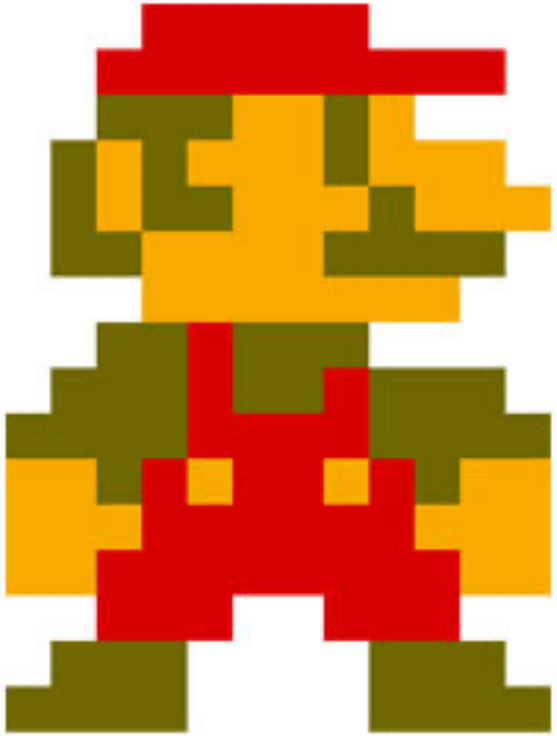
Sampling and quantization



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Source: Super Mario Games

Visual perception of the image

- Psycho-physical parameters: contrast, border, shape, texture, color...
- Human perception of images is prone to many **illusions**.

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Noise

- Real images are often degraded by some random errors. This is the **noise**.
- **White noise**. All frequencies are present with same intensity.
- **Gaussian noise**.
- The signal-independent degradation is called **additiv noise**.

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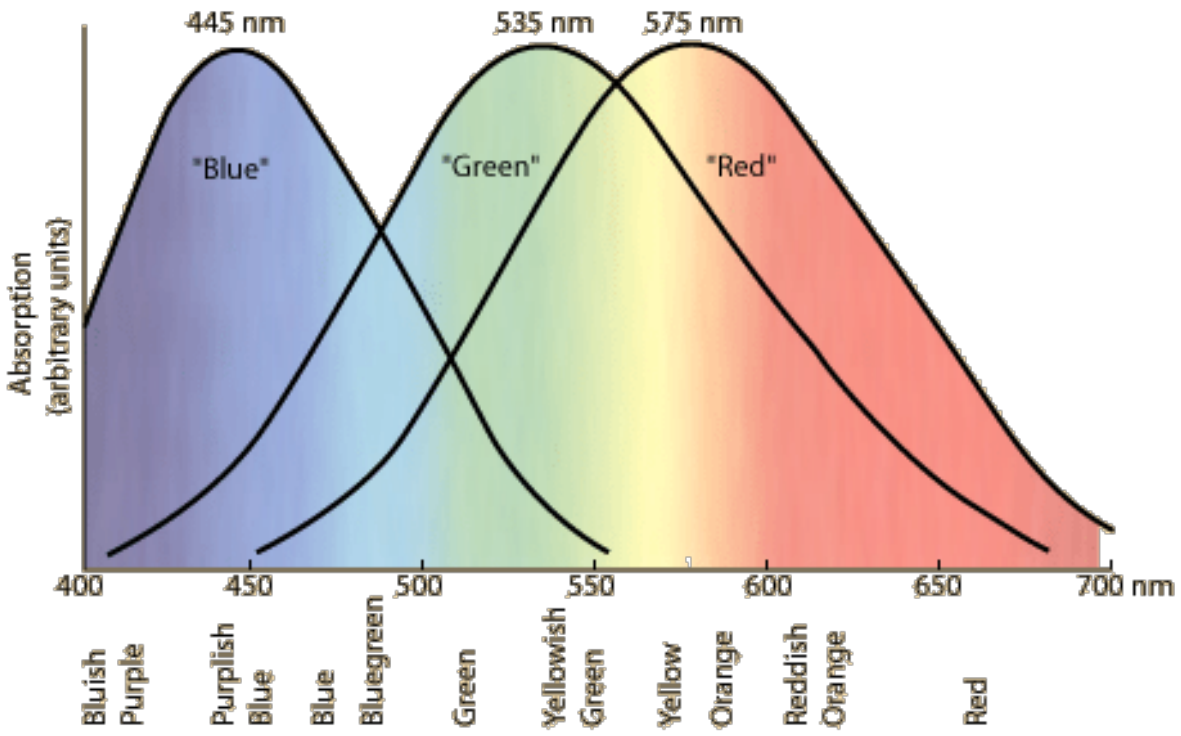
Human color perception



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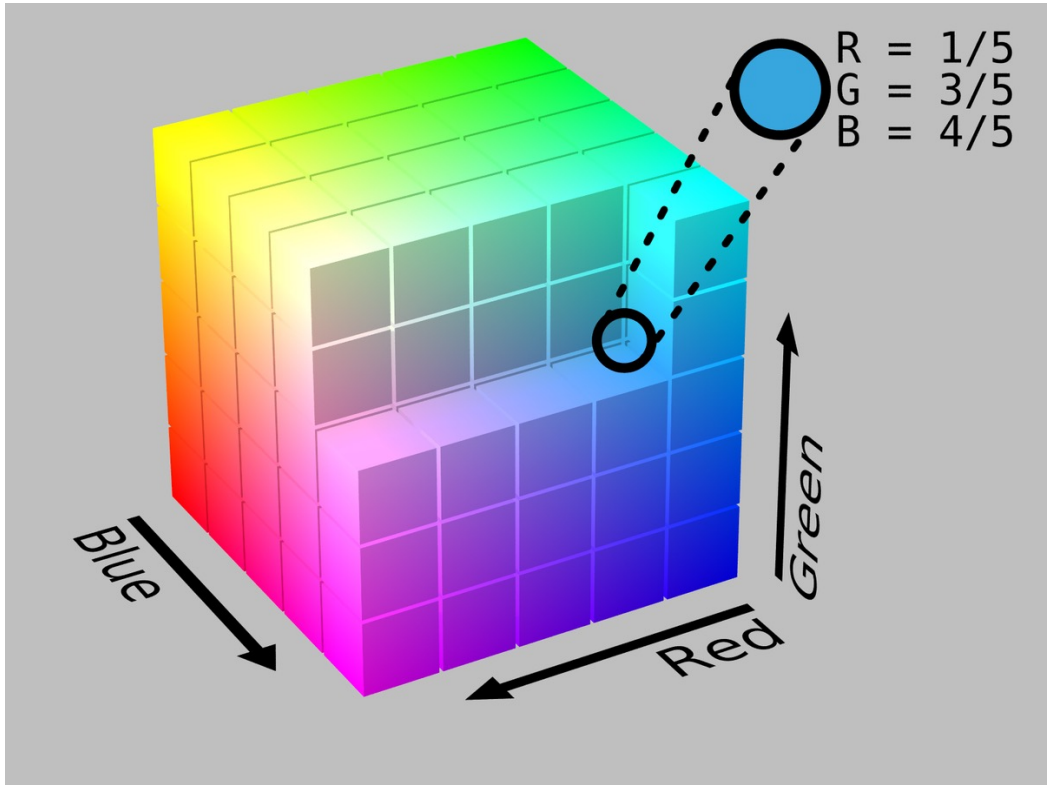
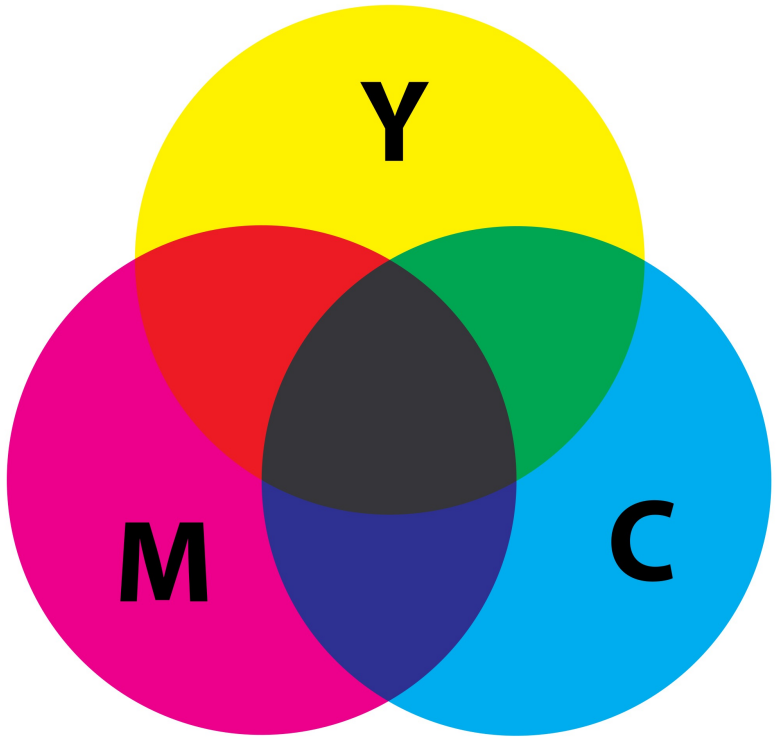
Source: http://hyperphysics.phy-astr.gsu.edu/hbase/vision/colcon.html

Color spaces - RGB



Source: https://en.wikipedia.org/wiki/HSL_and_HSV

Color spaces - CMY



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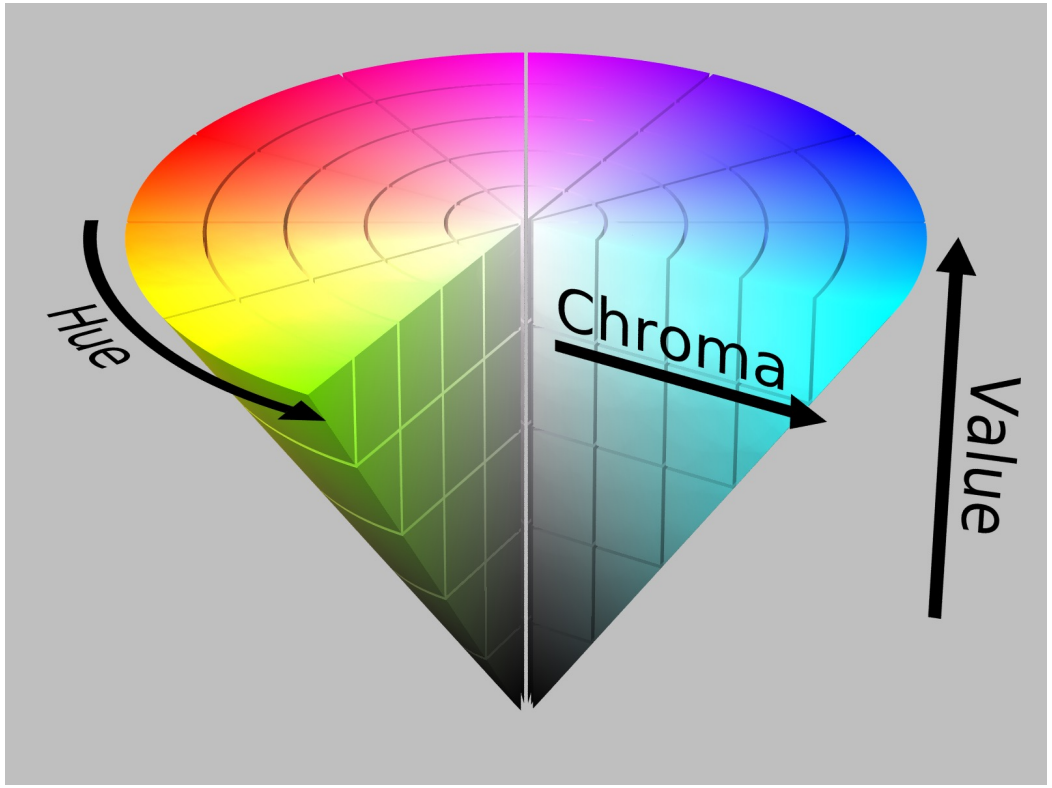
Color spaces - HSV



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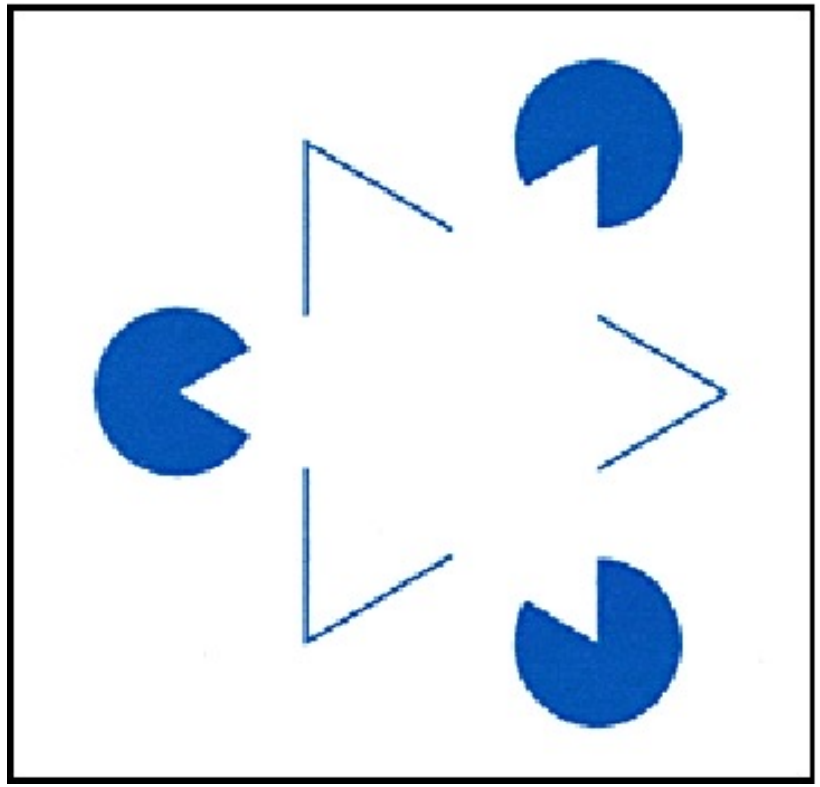
Source: https://en.wikipedia.org/wiki/HSL_and_HSV

Illusions



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Source: Unknown

Illusions

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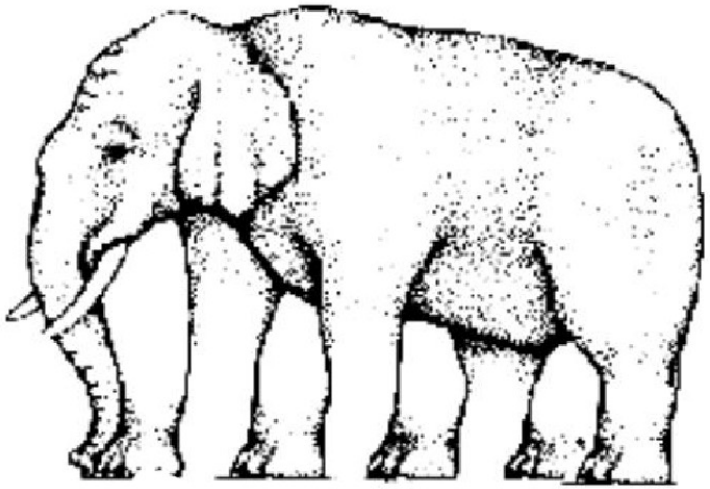
Source: http://www.fortresscomm.hu/varnagy/olvaso/optikai.csalodasok.htm

Illusions



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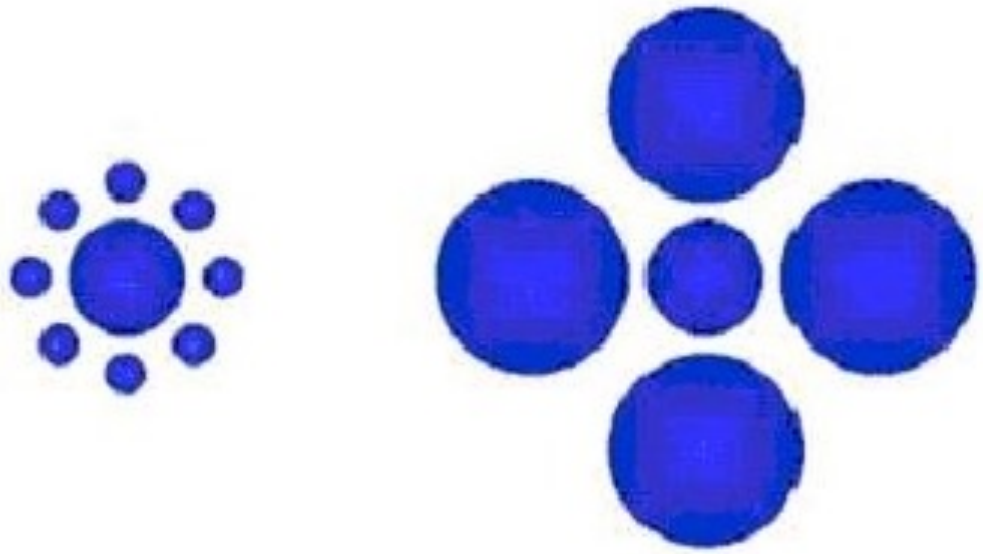
Sourse: Unknown

Illusions



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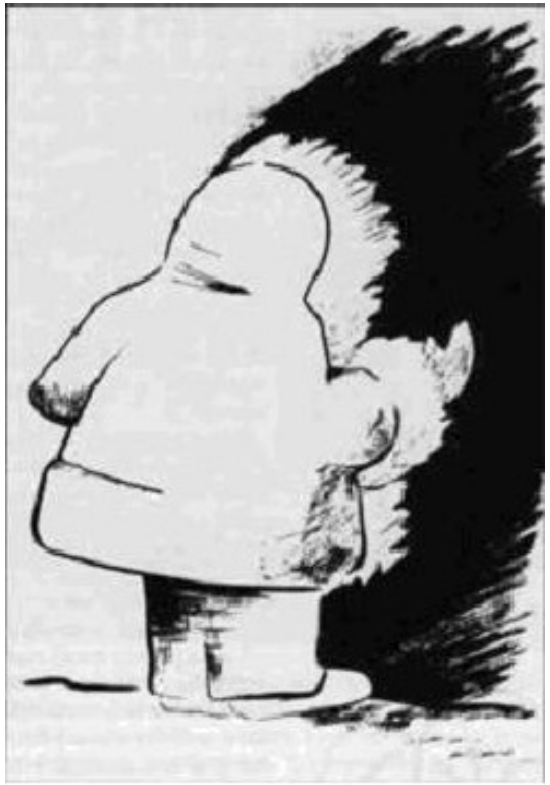
Illusions



Source: https://www.illusionsindex.org/i/eskimo-face

Mathematical background

Definition. The set $\boldsymbol{Z}^n$ is the n-dimensional digital plane, its elements are points. A non-empty and finite subset of the **digital plane** is *n*-dimensional **digital set**. Let $f : X \rightarrow \{0,1,...,m-1\} (m \in N)$ be a function over the digital set X. This function can be considered as a **digital picture** with *m* graylevels.

The pair $(p,f(p))$ $(p \in X)$ is **pixel** (picture element), where *p* is the **coords** and $f(p)$ is the **graylevel**.

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Mathematical background

Definition. Let f be a digital picture over a given digital set X. The digital picture f with two graylevels is a **binary picture**. The set $A\ F = \{p \in X | f(p) = 1\}$ is the **foreground**, and the set $B = \{p \in X \mid f(p) = 0\}$ is the **background** of the picture. The elements of the set F are the **object points** and the elements of the set B are the **background points**.

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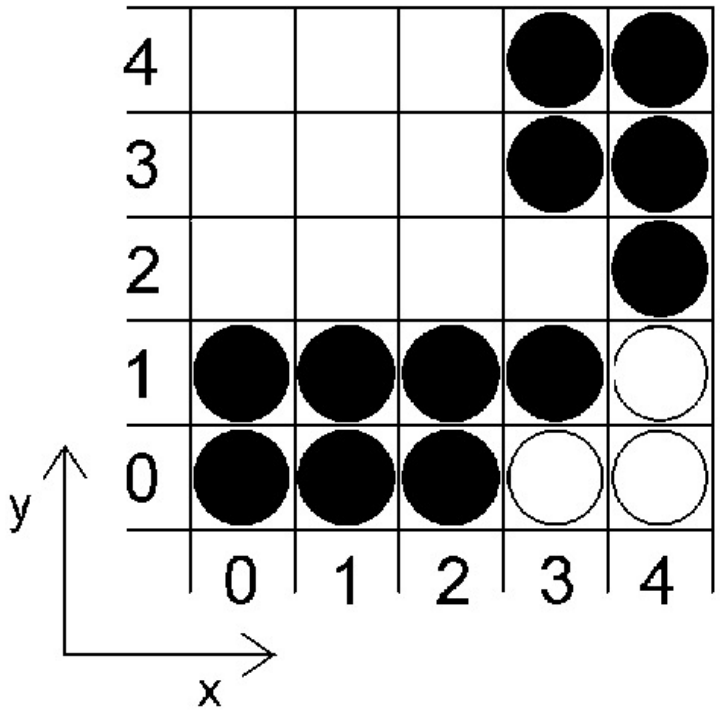
Example



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Example

$X = \{(0, 0), (1, 0), (2, 0), (3, 0), (4, 0), (0, 1), (1, 1), (2, 1), (3, 1), (4, 1), (4, 2), (3, 3), (4, 3), (3, 4), (4, 4)\}$.

$f(p)=1$, if $|x-y|\leq 2$, 0 othercase.

$F = \{(0, 0), (1, 0), (2, 0), (0, 1), (1, 1), (2, 1), (3, 1), (4, 2), (3, 3), (4, 3), (3, 4), (4, 4)\}$.

$B = X \setminus F = \{(3,0),(4,0),(4,1)\}$.

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Thank you for your attention